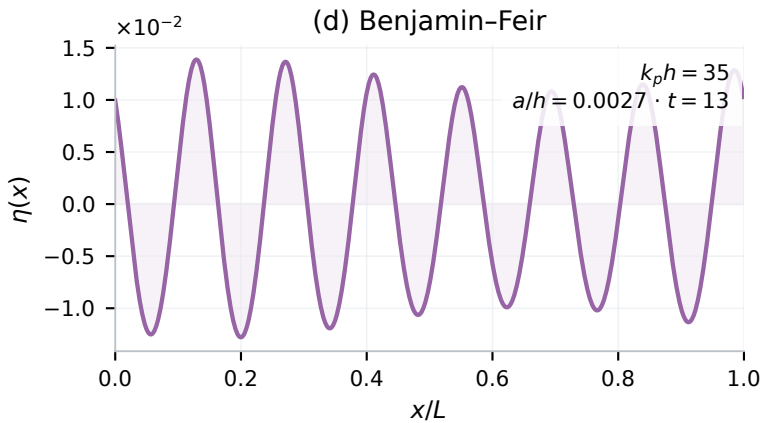
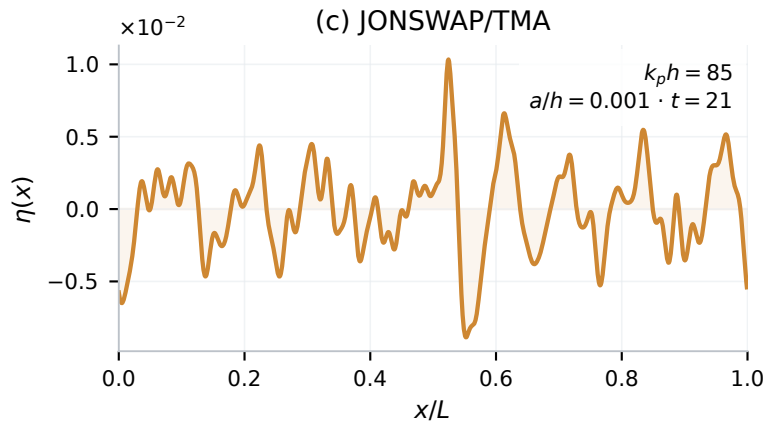
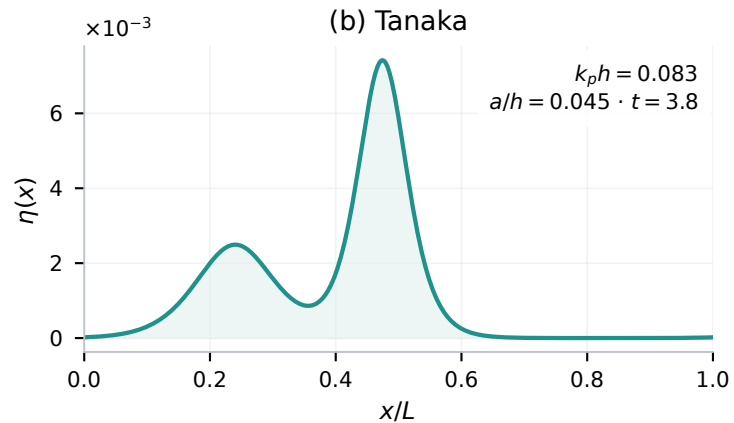
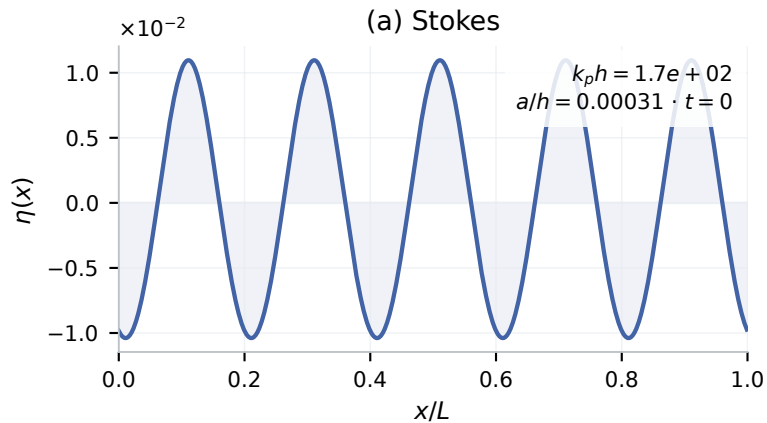
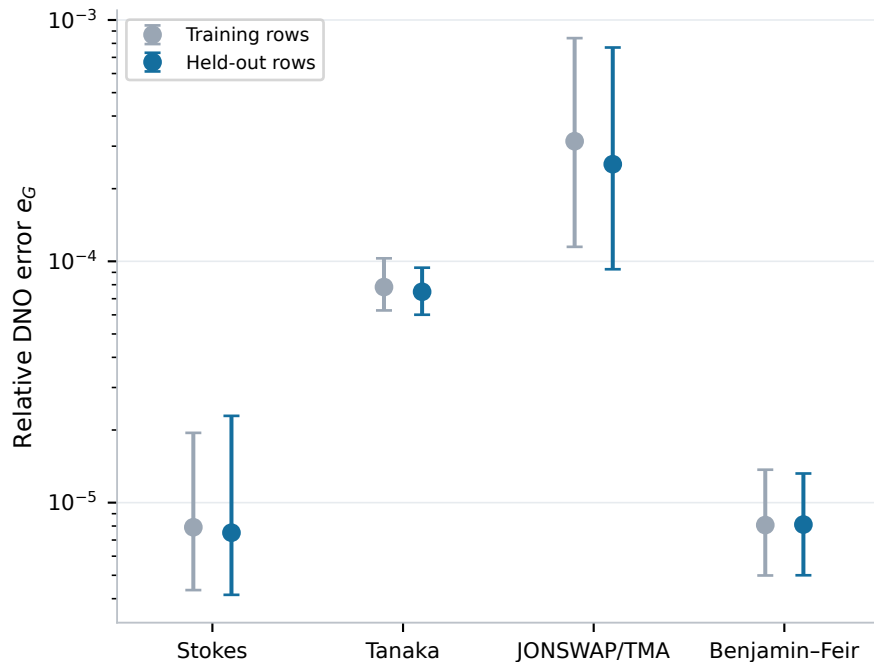
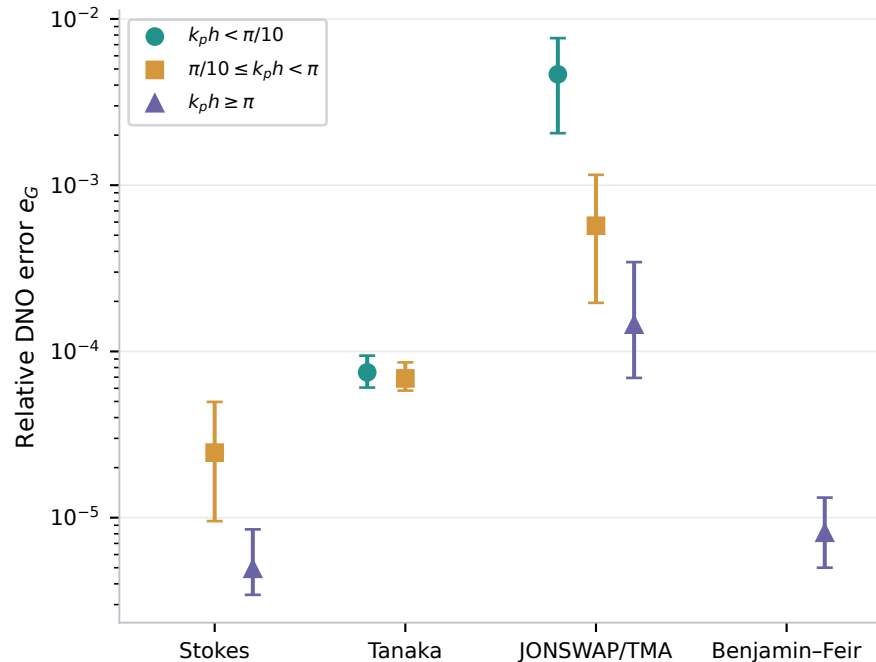




(a) Simulation-split accuracy

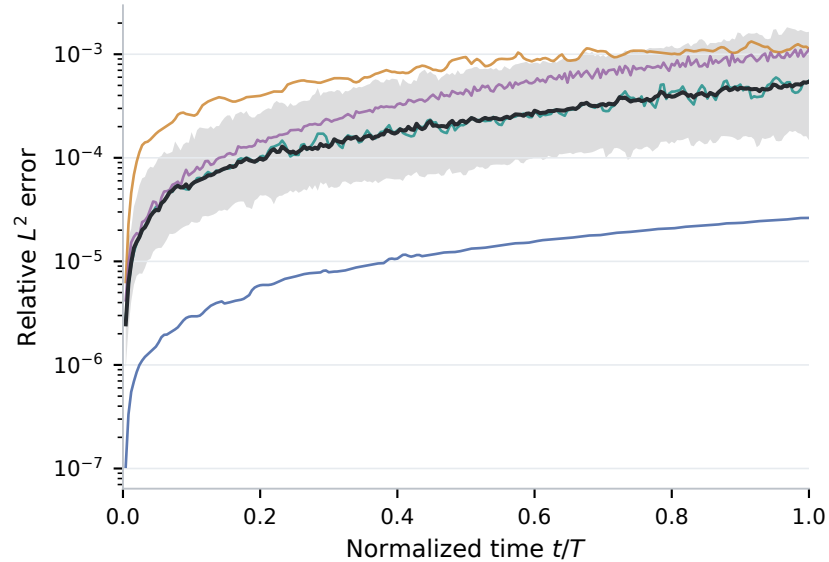
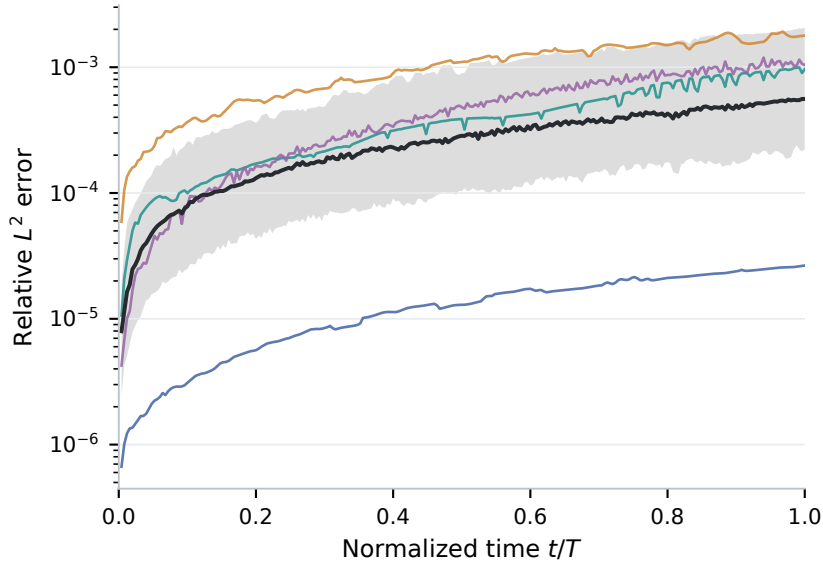


(b) Test rows by depth

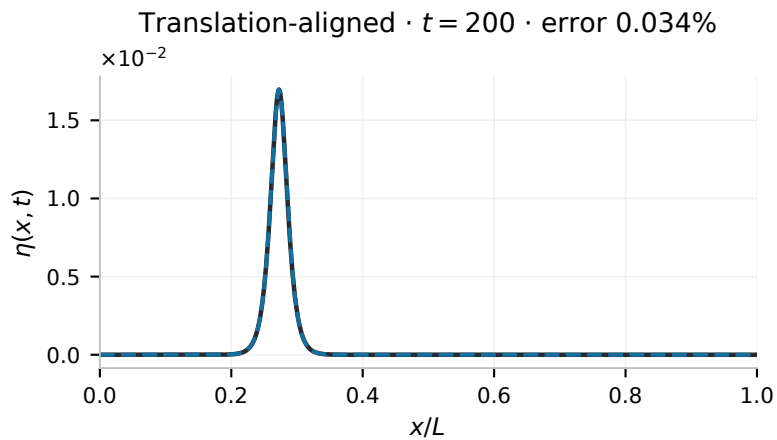
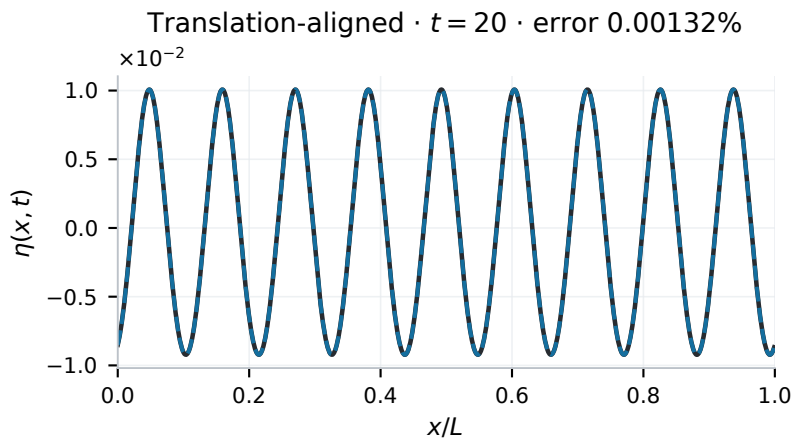
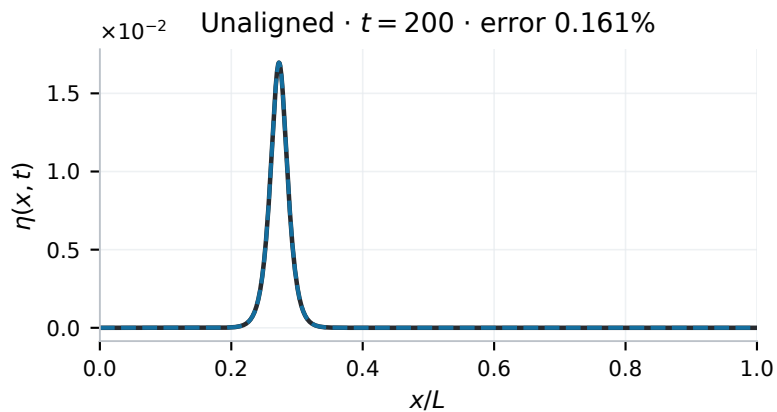
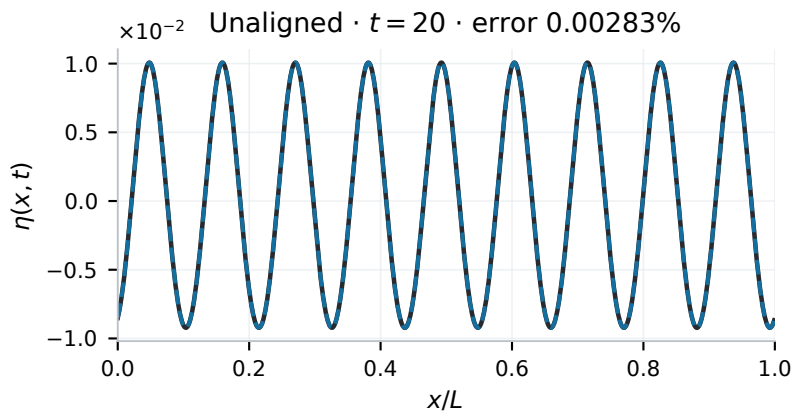
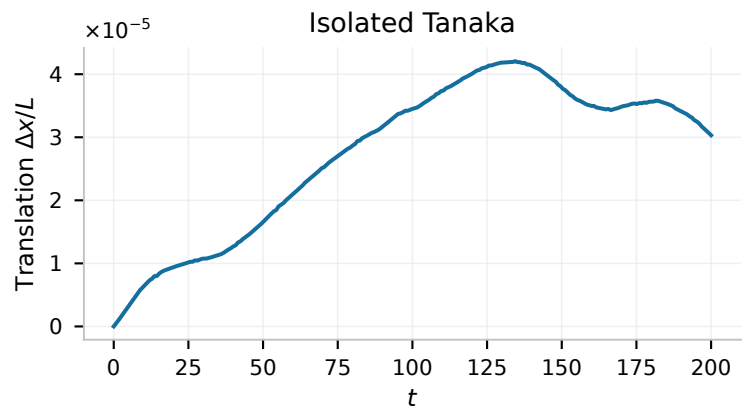
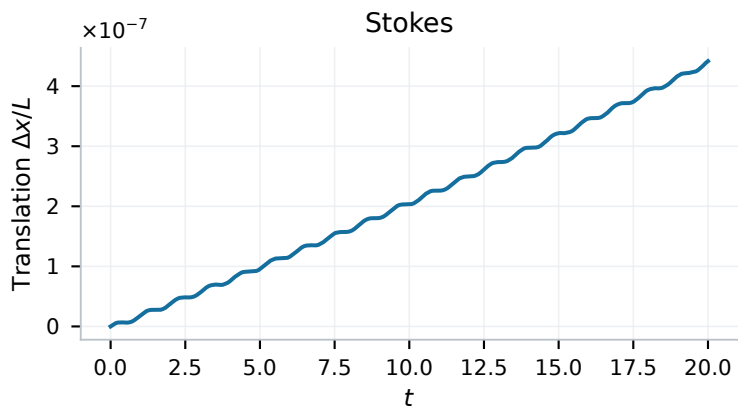


— Stokes — Tanaka — JONSWAP/TMA — Benjamin-Feir — Pooled learned DNO

(a) Surface elevation $e_\eta(t)$ (b) Potential $e_\xi(t)$



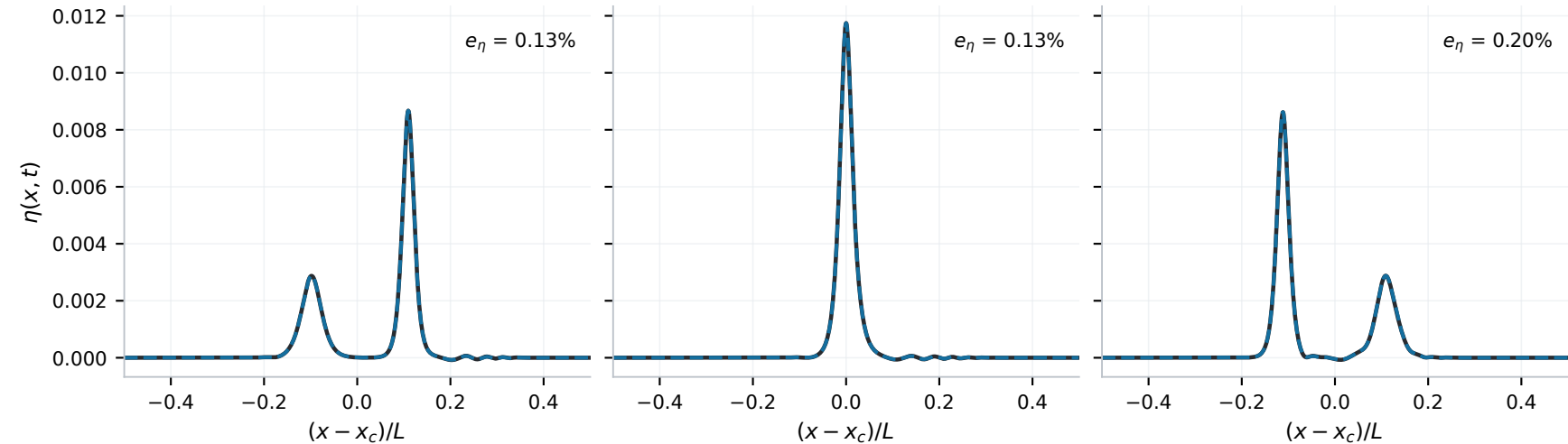
— CS reference - - - Learned DNO



— CS reference — Learned DNO
Interaction · $t = 17.6$

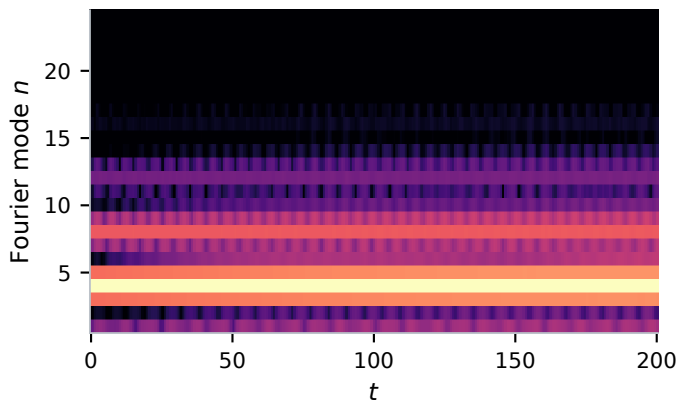
Before · $t = 14.4$

After · $t = 20.8$

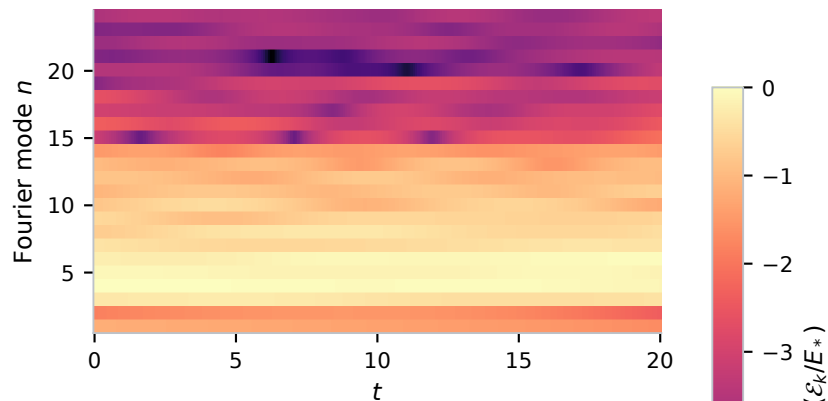


Two-wave Tanaka interaction · fixed recentering of both methods · event selected from reference crest amplification

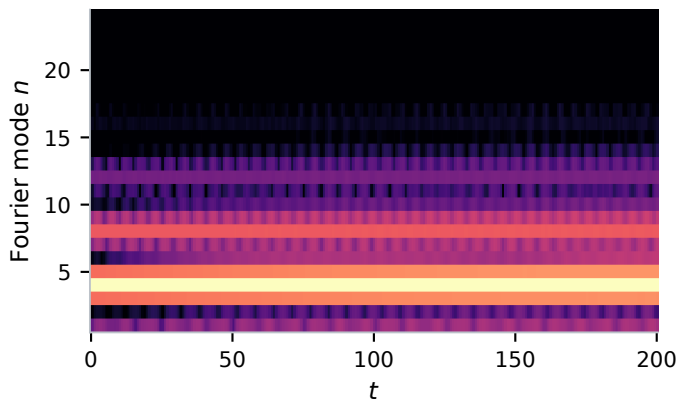
Benjamin-Feir · CS reference



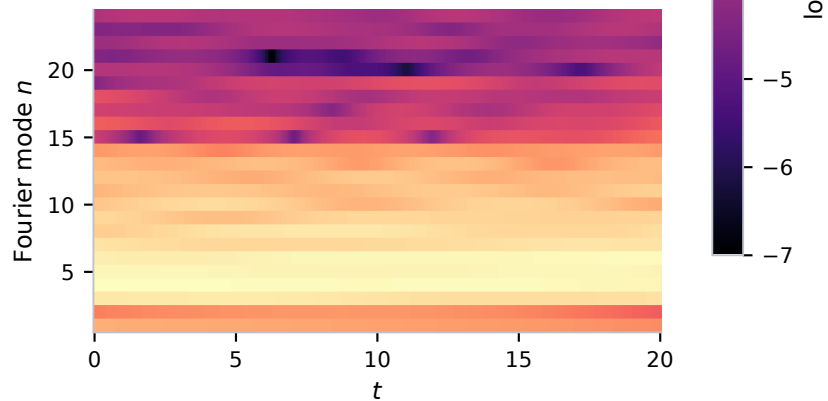
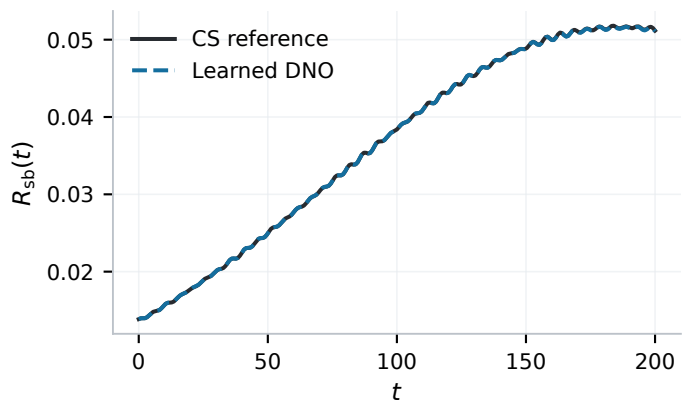
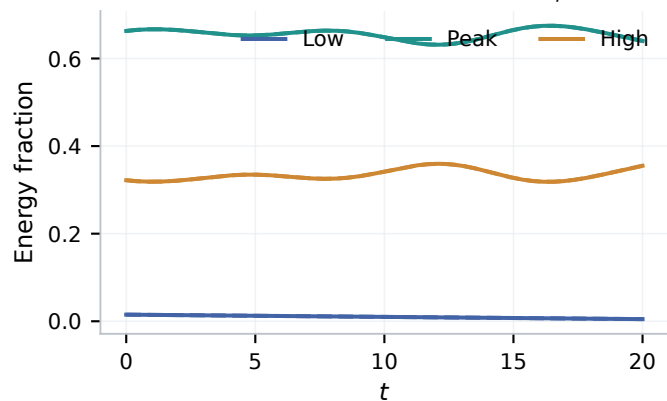
JONSWAP/TMA · CS reference



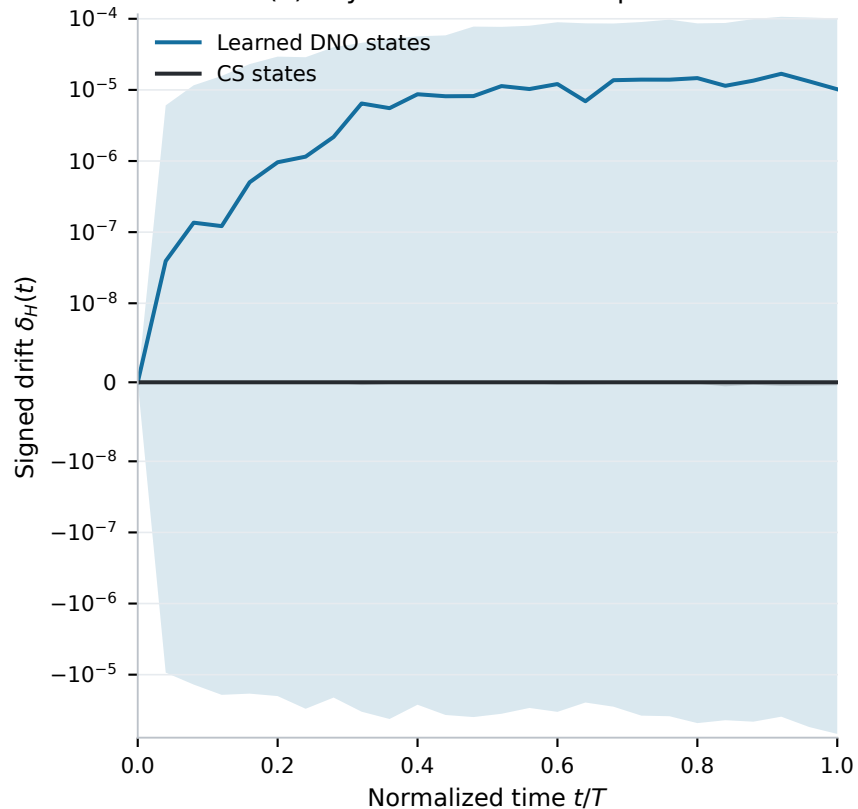
Benjamin-Feir · Learned DNO



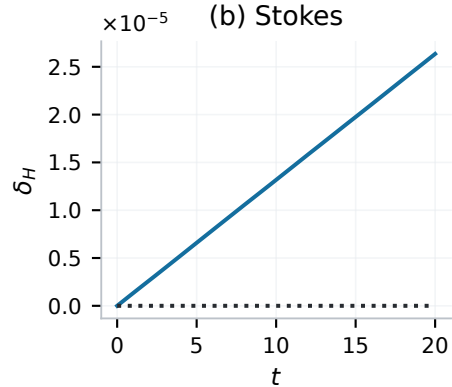
JONSWAP/TMA · Learned DNO

Seeded sidebands: $n_0 = 4, q = 1$ Bands relative to initial peak $n_p = 4$ 

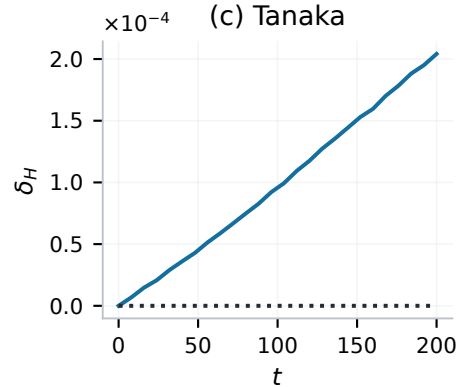
(a) Physical Hamiltonian · pooled



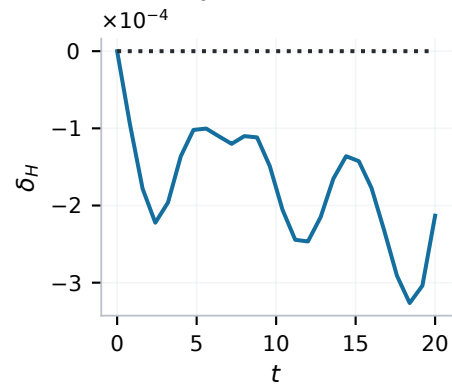
(b) Stokes



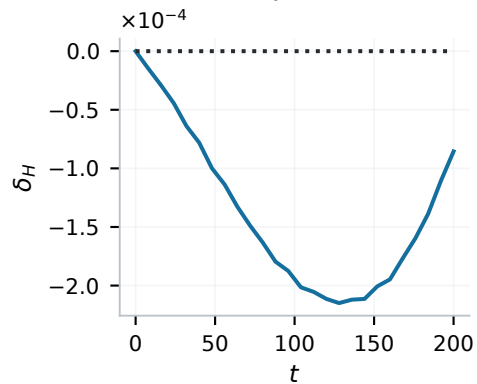
(c) Tanaka



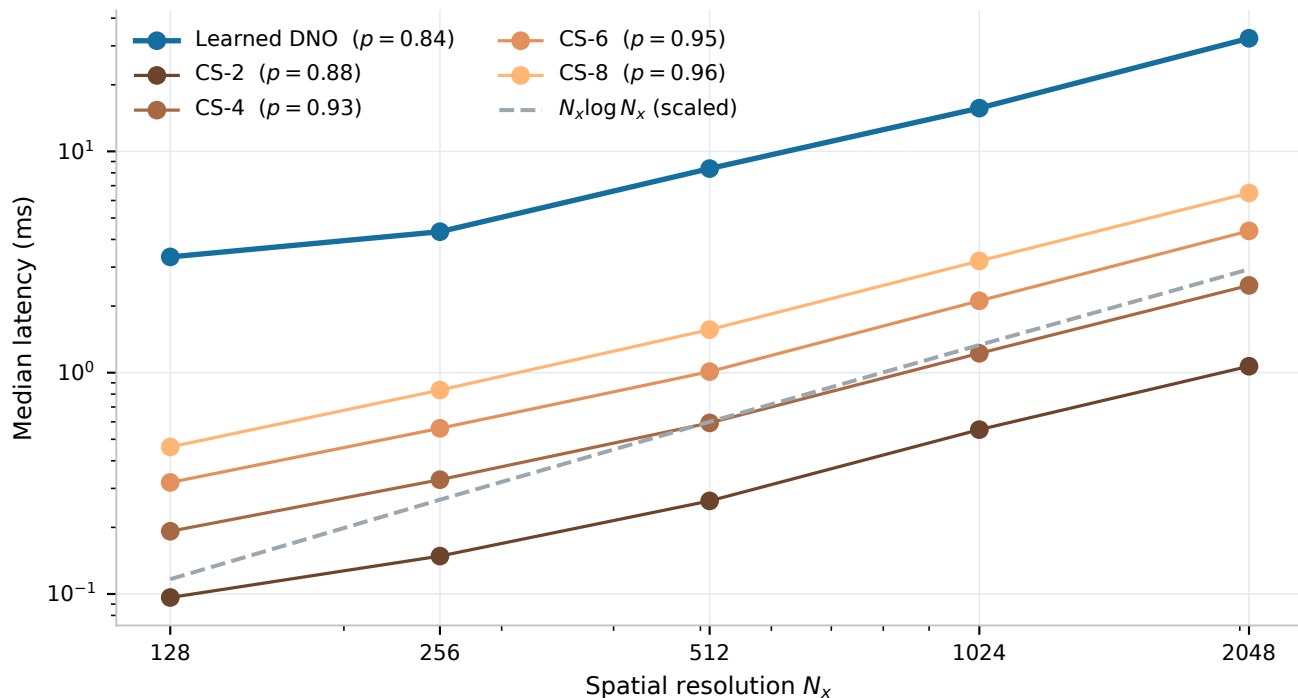
(d) JONSWAP/TMA



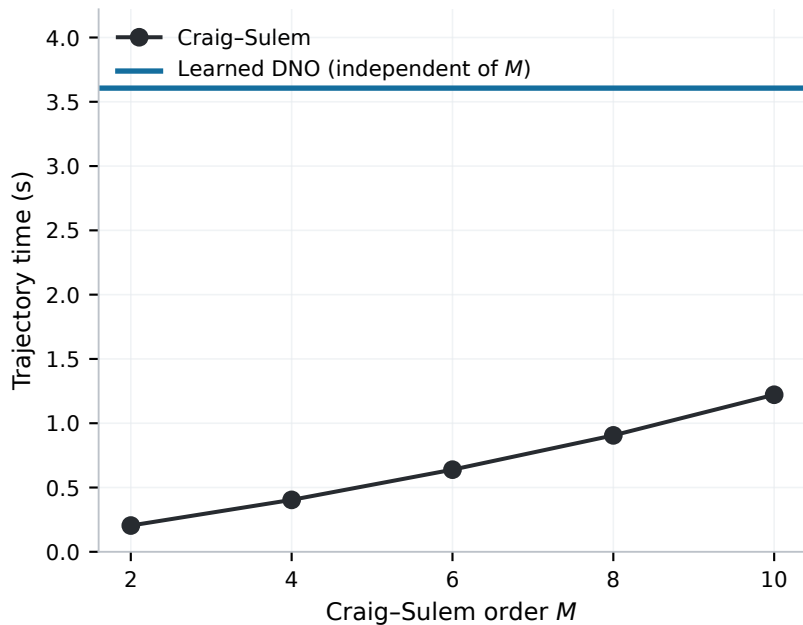
(e) Benjamin-Feir



Single DNO evaluation · warmed CPU execution



(a) Fixed-horizon runtime



(b) Accuracy-runtime comparison

